Troy Mayer

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Open to Summer 2027 internships in robot software and controls

EDUCATION

Purdue University, West Lafayette, IN Expected May 2028
B.S. Mechanical Engineering and B.S. Applied Statistics, Minor in Economics GPA: 3.32/4.00

SKILLS

- **Languages:** C++, Python, MATLAB, SQL
- **Robotics:** state estimation (Kalman filtering), PI and feed-forward control, QP force control, rigid-body kinematics, MuJoCo, Eigen, FreeRTOS
- **Tools:** Linux, Git, CMake, GoogleTest, GitHub Actions CI, Flask, pandas, PyTorch, KiCad, LabVIEW, CATIA, Creo

ROBOTICS EXPERIENCE AND PROJECTS

Boston Dynamics Jul 2026 – Present
Sensors and Perception Co-Op (Atlas) Waltham, MA

- Mapping impacts onto Atlas's mesh with encoders and **forward kinematics**, estimating impact energy from Khatib's operational-space effective mass and an OLS velocity fit; building a physics-based MuJoCo impact simulation.
- Building an impact forensics pipeline and Flask/Plotly dashboard on a Linux VM over ~28k impact events from ~900 test logs: ETL into SQLite, a 3D impact heatmap, and per-event waveforms used by engineers during testing.
- Modeling hardware thermal behavior in Python, using Savitzky-Golay filtering and multi-start least squares to predict steady-state temperatures across hundreds of units without full-length tests.

Legged Robot State Estimation and Control, *personal project, code* Aug 2026 – Oct 2026

- Built a **contact-aided error-state Kalman filter in C++** for a simulated Unitree Go1, fusing IMU and leg kinematics to 0.013 m/s velocity error, against 1.38 m/s for the IMU alone.
- Verified it with GoogleTest (finite-difference Jacobian checks, parity with a Python reference) and an observability analysis of what the sensors can and can't see.
- Stress-tested it under foot slip and contact latency, found tilt only weakly observable on straight walks, and fixed the tilt and latency failures.
- Closed the loop in Python simulation: the walking controller runs on the estimate without a fall, plus an MIT Cheetah 3-style QP balance controller.

Purdue ASME Biomechanics, Active Orthosis Apr 2024 – May 2026
Electronics Team Lead West Lafayette, IN

- Designed a **1 kHz PI motor velocity controller in C++** for an ESP32-S3, tuned from a motor model with feed-forward and anti-windup, and verified it in co-simulation with a model of the motor and forearm.
- Wrote the FreeRTOS firmware with fault monitors, a watchdog, and a kill switch, tested with GoogleTest in CI.
- Cut simulated EMG-to-motion latency from 103 to 43 ms on real EMG via new signal processing and loop tuning.
- Led the 5-engineer electronics team and redesigned the EMG board schematic in KiCad, raising cross-day gesture decoding from 31% to 87% in emulation.

Asteroid Hopping Robot, *independent research with Purdue University and UC3M (Madrid)* Oct 2026 – Present

- Building a 3-legged hopping robot and landing rig; designing push-off, landing, and mid-air attitude control in MuJoCo.

ADDITIONAL EXPERIENCE

Caterpillar (via Purdue Data Mine), *Machine Learning Researcher* Jan 2026 – May 2026

- Built LSTM/XGBoost forecasts of daily paint use with Monte Carlo confidence intervals, contributing to 4.7% less material use.

Chortos Lab, Purdue, *Undergraduate Researcher* Aug 2025 – May 2026

- Wrote a Python computer-vision tracker and LabVIEW/MATLAB data acquisition to measure gel samples, and used them to formulate gels with over 50% more force output.

Emerson, Discrete Automation (ASCO), *Mechanical and Manufacturing Engineer Intern* Jun 2025 – Aug 2025

- Designed and CNC-machined a spring canister for an industrial rotary valve in Creo with GD&T drawings; ran UL 429 validation on solenoid valves.